

Continued Unit - II on Statistics ...

Introduction: Statistics is a branch of mathematics and it is a study of the collection of a data, Organization of the data, Presentation of the data, analysis of the data and interpretation of the data. The statistics is very much useful in government organizations for the analysis of the data and also private organizations. In the present unit we discuss the some topics in the statistics like curve fitting of a given data using least squares method, correlation and regressions of the given data. Multiple correlation and regression for the given data.

Some Basic Formulas:

Let $x_0, x_1, x_2, \dots, x_n$ are the set of n values of the x and $y_0, y_1, y_2, \dots, y_m$ are the set of m values of the y , then

The mean value or average of x is $\bar{x}$ and is given by

$$\bar{x} = \frac{\sum_{i=0}^n x_i}{n} = \frac{\sum x}{n}$$

The variance of the x is σ_x^2 and is given by

$$\sigma_x^2 = \frac{\sum_{i=0}^n (x_i - \bar{x})^2}{n} = \frac{\sum (x - \bar{x})^2}{n}$$

The alternative formula for variance is

$$\sigma_x^2 = \frac{\sum x^2}{n} - (\bar{x})^2$$

Continued Unit - II on Statistics ...

The standard deviation of x is σ_x is positive square root of the variance of x and is given by

$$\sigma_x = \sqrt{\text{Variance}} = \sqrt{V(x)}$$

Similarly, for the y The mean value or average of y is $\bar{y}$ and is given by

$$\bar{y} = \frac{\sum_{i=0}^m y_i}{m} = \frac{\sum y}{m}$$

The variance of the y is σ_y^2 and is given by

$$\sigma_y^2 = \frac{\sum_{i=0}^m (y_i - \bar{y})^2}{m} = \frac{\sum (y - \bar{y})^2}{m}$$

The alternative formula for variance is

$$\sigma_y^2 = \frac{\sum y^2}{m} - (\bar{y})^2$$

The standard deviation of y is σ_y is positive square root of the variance of y and is given by

$$\sigma_y = \sqrt{\text{Variance}} = \sqrt{V(y)}$$

Continued Unit - II on Curve fitting ...

Curve Fitting:- The process of obtaining an approximate relation between two variables x and y from the given values of x and y is called a curve fitting. We discuss the curve fitting like, curve fitting of the straight line, curve fitting of the parabola, curve fitting of a exponential curves and curve fitting of a geometric curves. The curve fitting of the given data is obtained by using **The Method of Least Squares**.

Estimated Errors (OR) Residuals:-

The difference between the observed values of x i.e. (y) and estimated values of x i.e. (Y) is called as a **Estimated errors (OR) Residuals**. The estimated errors is denoted by R_i or R and is defined as

$$R_i = (y_i - Y_i) \text{ (OR) } R = (y - Y)$$

Method of Least Squares:

Statement: The method of least squares states that the sum of the squares of the estimated errors is minimum.

The method of least squares is denoted by S and mathematically define as

$$S = \sum_{i=0}^n R_i^2 = \sum_{i=0}^n (y_i - Y_i)^2$$

(OR)

$$S = \sum R^2 = \sum (y - Y)^2$$

Continued Unit - II on Curve Fitting of a Straight Line ...

Curve Fitting of a Straight Line (OR) Curve Fitting of a Linear Curves:
Obtain the normal equations to fit a straight line of the form $y = ax + b$.

Consider a straight line of the form $y = ax + b$, where a and b are two parameters. To find an approximate values of the parameters a and b using the given set of values of x and y . Substituting the values of a and b in the straight line $y = ax + b$, we can get an approximate equation of the straight line for the given data.

We have by the method of least squares

$$S = \sum R^2 = \sum (y - Y)^2$$
$$S = \sum [y - (ax + b)]^2 \quad (1)$$

From this we can observed that S is a function of two parameters a and b . The necessary condition for S to be minimum(least) if $\frac{\partial S}{\partial a} = 0$ and $\frac{\partial S}{\partial b} = 0$.

Now, differentiate equation (1) partially w.r.t. a and equating to zero, we get

$$\frac{\partial S}{\partial a} = 0$$

→

$$2 \sum [y - (ax + b)](-x) = 0$$

Dividing through out by 2 and rearrange

$$\sum [-xy + ax^2 + bx] = 0$$

Continued Unit - II on Curve Fitting of a Straight Line ...

$$\begin{aligned} -\sum xy + a \sum x^2 + b \sum x &= 0 \\ a \sum x^2 + b \sum x &= \sum xy \end{aligned} \quad (2)$$

Again differentiate equation (1) partially w.r.t. b and equating to zero, we get

$$\frac{\partial S}{\partial b} = 0$$

→

$$2 \sum [y - (ax + b)](-1) = 0$$

Dividing through out by 2 and rearrange

$$\begin{aligned} \sum [-xy + ax + b] &= 0 \\ -\sum y + a \sum x + \sum b &= 0 \end{aligned}$$

But, $\sum b$ means sum b , n times that is $\sum b = nb$

$$\begin{aligned} -\sum y + a \sum x + nb &= 0 \\ a \sum x + nb &= \sum y \end{aligned} \quad (3)$$

Continued Unit - II on Curve Fitting of a Straight Line ...

Therefore, equations (2) and (3) are called as normal equations for the curve fitting of the straight line of the form $y = ax + b$.

Similarly, we can find an approximate straight line of the form $y = a + bx$ and the corresponding normal equations are

$$\begin{aligned} na + b \sum x &= \sum y \\ a \sum x + b \sum x^2 &= \sum xy \end{aligned}$$

Note: If the given curve is not in standard form the straight line by rewriting it to standard form by using some basic simplifications.

Working process to solve the problems:

Step - 01: Write the relevant normal equations for the straight line.

Step - 02: Construct the relevant table according to the normal equations from the given data.

Step - 03: Substituting all table values in the normal equations, we get two simultaneous equations in two variables a and b , by solving these equations by using elimination method find the values of a and b .

Step - 04: Substituting the values of a and b in the relevant straight line.

Continued Unit - II Problems on Curve Fitting of a Straight Line ...

Note: Suppose the equation of the straight line is not given then we can assume any one of the form $y = a + bx$ (OR) $y = ax + b$.

Problems:

Problem - 01: Fit an approximate straight line, by using the method of least - squares from the following data. Hence find the approximate value of y at $x = 1.5$

x	1	2	3	4	5
y	14	13	9	5	2

Solution: Let us consider a straight line of the form.

$$y = a + bx \quad (4)$$

We know that, the normal equations for straight line are

$$na + b \sum x = \sum y \quad (5)$$

$$a \sum x + b \sum x^2 = \sum xy \quad (6)$$

Here, number of observation are given $n = 5$ and construct the table according to the normal equations

Continued Unit - II Problems on Curve Fitting of a Straight Line ...

x	y	x^2	xy
1	14	1	14
2	13	4	26
3	9	9	27
4	5	16	20
5	2	25	10
$\sum x = 15$	$\sum y = 43$	$\sum x^2 = 55$	$\sum xy = 97$

Substituting the values of $n = 5$, $\sum x = 15$, $\sum y = 43$, $\sum x^2 = 55$ and $\sum xy = 97$ in the equations (5) and (6)

$$5a + 15b = 43 \quad (7)$$

$$15a + 55b = 97 \quad (8)$$

By solving the equations (7) and (8) by using elimination method, we get $a = 18.2$ and $b = -3.2$. Substituting the values of $a = 18.2$ and $b = -3.2$ in equation (4). Hence the required equation of the straight for the given data is

$$y = 18.2 - 3.2x$$

$\therefore$ the value of y at $x = 1.5 \rightarrow y = 13.4$.

Continued Unit - II Problems on Curve Fitting of a Straight Line...

Problem - 02: Determine the best - fitting straight line for the following data, using the method of least - squares.

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

Solution: Let us consider a straight line of the form.

$$y = a + bx \quad (9)$$

We know that, the normal equations for the straight line are

$$na + b \sum x = \sum y \quad (10)$$

$$a \sum x + b \sum x^2 = \sum xy \quad (11)$$

Here, the number of observation are given $n = 8$ and construct the table according to the normal equations

Continued Unit - II Problems on Curve Fitting of a Straight Line ...

x	y	x^2	xy
1	1	1	1
3	2	9	6
4	4	16	16
6	4	36	24
8	5	64	40
9	7	81	63
11	8	121	88
14	9	196	126
$\sum x = 56$	$\sum y = 40$	$\sum x^2 = 524$	$\sum xy = 364$

Substituting the values of $n = 8$, $\sum x = 56$, $\sum y = 40$, $\sum x^2 = 524$ and $\sum xy = 364$ in the equations (10) and (11)

$$8a + 56b = 40 \quad (12)$$

$$56a + 524b = 364 \quad (13)$$

By solving the equations (12) and (13) using elimination method, we get $a = \frac{18}{33} = 0.545454$ and $b = \frac{21}{33} = 0.363636$. Substituting the values of a and b in equation (15). Hence the required equation of the straight for the given data is

$$y = 0.545454 + 0.363636x$$

Continued Unit - II Problems on Curve Fitting of a Straight Line ...

Problem - 03: Find a law of the form $V = a + \frac{b}{A}$ for the following data using the method of least - squares. Hence find an approximate value of V at $A = 12$

A	2	3	4	6	10
V	50	47	46	45	44

Solution: By data, we need to find a a straight line of the form.

$$V = a + \frac{b}{A}$$

But it is not in standard form converting to standard form by taking $X = \frac{1}{A}$

$$V = a + bX$$

We know that, the normal equations for straight line are

$$na + b \sum X = \sum y \quad (14)$$

$$a \sum X + b \sum X^2 = \sum Xy \quad (15)$$

Here, number of observation are given $n = 5$ and construct the table according to the normal equations

Continued Unit - II Problems on Curve Fitting of a Straight Line ...

A	V	$X = \frac{1}{A}$	X^2	XV
2	50	0.5	0.25	25
3	47	0.3334	0.1112	16.6667
4	46	0.25	0.0625	11.5
6	45	0.16667	0.027778	7.5
10	44	0.1	0.01	4.4
	$\sum V = 232$	$\sum X = 1.35007$	$\sum X^2 = 0.461478$	$\sum XV = 65.0667$

Problem - 03: Using the method of least - squares, Fit a linear relation of the form $P = a + bW$ to the following data, where P is the pull required to fit a weight W . Hence estimate an approximate value of P when $W = 150kg$.

$W(Kg.wt.)$	50	70	100	120
$P(Kg.wt)$	12	15	21	25

Solution: By data, we need to find a a straight line of the form

$$P = a + bW$$

Continued Unit - II on curve Fitting of a Exponential Curves ...

Curve Fitting of a Exponential Curves:

Consider a exponential curve of the form $y = ae^{bx}$, where a and b are two parameters. To find an approximate values of the parameters a and b using the given set of values of x and y . Substituting the values of a and b in the exponential curve $y = ae^{bx}$, we can get an approximate equation of the exponential curve for the given data.

$$y = ae^{bx}$$

Applying logarithms on both sides and make use the properties of logarithms, we get

$$\log y = \log(ae^{bx})$$

$$\log y = \log a + \log e^{bx}$$

$$\log y = \log a + bx \log e$$

$$\text{but, } \log e = \log_e e = 1$$

$$\log y = \log a + bx$$

$$\text{Let } Y = \log y \text{ and } A = \log a \rightarrow a = e^A$$

$$Y = A + bx$$

Continued Unit - II on Curve Fitting of a Exponential Curves ...

Therefore, this is of the form straight line and can be solve by using concept of curve fitting of a straight line, find the values of A, b and $a = e^A$. Substituting the values of a and b in $y = ae^{bx}$, we get an exponential curve for the given data.

Note: The exponential curves are converted to straight line form by taking logarithm on both the sides and find the exponential curve for the given data by using the procedure of fitting a straight line.

Problems:

Problem - 01: Fit a curve of the form $y = ae^{bx}$ for the given below.

x	0	2	4
y	8.12	10	31.82

Solution: Given an exponential curve

$$y = ae^{bx} \quad (16)$$

Applying the logarithm on both sides and make use the properties of logarithms,

$$\log(y) = \log(ae^{bx})$$

$$\log(y) = \log(a) + \log(e^{bx})$$

$$\log(y) = \log(a) + (bx) \log e$$

but $\log e = 1$, let $Y = \log(y)$ and $A = \log(a)$

$$Y = A + bx$$

Continued Unit - II Problems on Curve Fitting of a Exponential Curves

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This is of the straight line form and the normal equations are

$$nA + b \sum x = \sum Y$$

$$A \sum x + b \sum x^2 = \sum xY$$

Here, the number of observations are $n = 3$ and the relevant table is

x	y	$Y = \log_e(y)$	x^2	xY
0	8.12	2.09433	0	0
2	10	2.30258	4	4.60
4	31.82	3.46009	16	13.84
$\sum x = 6$		$\sum y = 7.85$	$\sum x^2 = 20$	$\sum xY = 18.44$

Substituting all the values in the normal equations

$$3A + 6b = 7.85$$

$$6A + 20b = 18.24$$

Solving these equations by using elimination method we get, $b = 0.3425$ and $A = 1.932 \rightarrow a = e^A = e^{1.932} = 6.903$.

Continued Unit - II Problems on curve Fitting of a Exponential Curves

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Substituting the values of a and b in $y = ae^{bx}$ and the required exponential curve is $y = 6.903e^{0.3425x}$.

Problem - 02: Find the equation of the best fitting curve in the form $y = ae^{bx}$ for the data.

x	5	6	7	8	9	10
y	133	55	23	7	2	2

Solution: Given an exponential curve

$$y = ae^{bx} \quad (17)$$

Applying the logarithm on both sides and make use the properties of logarithms,

$$\log(y) = \log(ae^{bx})$$

$$\log(y) = \log(a) + \log(e^{bx})$$

$$\log(y) = \log(a) + (bx) \log e$$

but $\log e = 1$, let $Y = \log(y)$ and $A = \log(a)$

$$Y = A + bx$$

Continued Unit - II on curve Fitting of a Geometric Curves ...

Curve Fitting of a Geometric Curves:

Consider a geometric curve of the form $y = ax^b$ (OR) $y = ab^x$, where a and b are two parameters. To find an approximate values of the parameters a and b using the given set of values of x and y . Substituting the values of a and b in the geometric curve $y = ax^b$, we can get an approximate equation of the geometric curve for the given data.

$$y = ax^b$$

Applying logarithms on both sides and make use the properties of logarithms, we get

$$\log y = \log(ax^b)$$

$$\log y = \log a + \log x^b$$

$$\log y = \log a + b \log x$$

Take, $Y = \log y$, $A = \log a$ and $X = \log x$, we get

$$Y = A + bX$$

Therefore, this is nothing but the straight line and can be solve by using the concept of curve fitting of a straight line for the given data. Find the values of A , $a = e^A$ and b , substituting the values of a and b in $y = ax^b$, we get geometric curve for the given data.

Continued Unit - II Problems on Curve Fitting of a Geometric Curves ...

Problems:

Problem - 01: Fit a curve of the form $y = ab^x$ for the following data, using the method of least - squares.

x	1	2	3	4	5	6	7	8
y	1.0	1.2	1.8	2.5	3.6	4.7	6.6	9.1

Solution: By data, we need to find curve of the form .

$$y = ab^x \quad (18)$$

Applying logarithms on both sides and make use the properties of logarithms, we get

$$\log y = \log(ab^x)$$

$$\log y = \log a + \log b^x$$

$$\log y = \log a + x \log b$$

Take, $Y = \log y$, $A = \log a$ and $B = \log b$, we get

$$Y = A + Bx$$

This is like a straight line and can be solve by using the procedure of the curve fitting of a straight line

Continued Unit - II Problems on Curve Fitting of a Geometric Curves ...

We know that, the normal equations are

$$nA + B \sum x = \sum Y$$

$$A \sum x + B \sum x^2 = \sum xY$$

Here, the number of observation are $n = 8$ and the table is

x	y	$Y = \log y$	x^2	xY
1	1.0	0	1	0
2	1.2	0.1823	4	0.3646
3	1.8	0.5878	9	1.7634
4	2.5	0.9163	16	3.6652
5	3.6	1.2809	25	6.4045
6	4.7	1.5476	36	9.2856
7	6.6	1.8871	49	13.20097
8	9.1	2.2083	64	17.6664
$\sum x = 36$	$\sum y = 30.5$	$\sum Y = 8.6103$	$\sum x^2 = 204$	$\sum xY = 52.3594$

Continued Unit - II Problems on Curve Fitting of a Geometric Curves ...

Substituting all these values in the normal equations, we get

$$8A + 36B = 8.6103$$

$$36A + 204B = 52.3694$$

Solving these equations by elimination method and find the values of $A = -0.3822$ and $B = 0.3241$

$$\rightarrow a = e^A = e^{-0.3822} = 0.6824 \text{ and}$$

$$b = e^B = e^{0.3241} = 1.3828$$

Substituting the values of a and b in $y = ab^x$ and hence the required curve for the given data is $y = (0.6824)(1.3828)^x$.

Problem - 01: Fit a curve of the form $y = ab^x$ for the following data, using the method of least - squares.

x	0	2	4	5	7	10
y	100	120	256	390	710	1600

Solution: By data, we need to find curve of the form .

$$y = ab^x \tag{19}$$

Continued Unit - II on curve Fitting of a second degree Parabola ...

Curve Fitting of a second degree Parabola (OR) Curve Fitting of a Quadratic Curves:

Consider a second degree parabola of the form $y = ax^2 + bx + c$, where a , b and c are three parameters. To find an approximate values of the parameters a , b and c using the given set of values of x and y . Substituting the values of a , b and c in the second degree parabola $y = ax^2 + bx + c$, we can get an approximate equation of the second degree parabola for the given data.

We have by the method of least squares

$$S = \sum R^2 = \sum (y - Y)^2$$
$$S = \sum [y - (ax^2 + bx + c)]^2 \quad (20)$$

From this we can observed that S is a function of three parameters a , b and c . The necessary condition for S to be minimum(least) if $\frac{\partial S}{\partial a} = 0$, $\frac{\partial S}{\partial b} = 0$ and $\frac{\partial S}{\partial c} = 0$.

Now, differentiate equation (10) partially w.r.t. a , b and c equating to zero, we get

$$\frac{\partial S}{\partial a} = 0$$

→

$$2 \sum [y - (ax^2 + bx + c)](-x^2) = 0$$

Dividing through out by 2 and rearrange

Continued Unit - II on Curve Fitting of a second degree Parabola ...

$$\begin{aligned} -\sum x^2 y + a \sum x^4 + b \sum x^3 + c \sum x^2 &= 0 \\ a \sum x^4 + b \sum x^3 + c \sum x^2 &= \sum x^2 y \\ \frac{\partial S}{\partial b} &= 0 \end{aligned} \tag{21}$$

→

$$2 \sum [y - (ax^2 + bx + c)](-x) = 0$$

Dividing through out by 2 and rearrange

$$\begin{aligned} \sum [-xy + ax^3 + bx^2 + cx] &= 0 \\ -\sum xy + a \sum x^3 + b \sum x^2 + c \sum x &= 0 \\ a \sum x^3 + b \sum x^2 + c \sum x &= \sum xy \\ \frac{\partial S}{\partial c} &= 0 \end{aligned} \tag{22}$$

→

$$2 \sum [y - (ax^2 + bx + c)](-1) = 0$$

Continued Unit - II on Curve Fitting of a second degree Parabola ...

Dividing through out by 2 and rearrange

$$\begin{aligned}\sum[-y + ax^2 + bx + c] &= 0 \\ -\sum y + a\sum x^2 + b\sum x + \sum c &= 0\end{aligned}$$

but, $\sum c = nc$

$$a\sum x^2 + b\sum x + nc = \sum y \quad (23)$$

Therefore, the equations (11), (12) and (13) are called as the normal equations of the curve fitting of a second degree parabola for the given data.

Note: If the given curve is not in standard form the second degree parabola by rewriting it to standard.

Working process to solve the problems:

Step - 01: Write the relevant normal equations for the second degree parabola.

Step - 02: Construct the relevant table according to the normal equations from the given data.

Step - 03: Substituting all table values in the normal equations, we get three simultaneous equations in three variables a , b and c , by solving these equations find the values of a , b and c .

Step - 04: Substituting the values of a , b and c in the relevant second degree parabola.

Continued Unit - II Problems on Curve Fitting of a second degree Parabola ...

Problems:

Problem - 01: Fit a parabola $y = a + bx + cx^2$ by the method of least - squares to the following data.

x	1	2	3	4	5	6	7
y	2.3	5.2	9.7	16.5	29.4	35.5	54.4

Solution: By data, we need to find the second degree parabola of the form

$$y = a + bx + cx^2 \quad (24)$$

We know that, the normal equations for the given parabola are

$$na + b \sum x + c \sum x^2 = \sum y \quad (25)$$

$$a \sum x + b \sum x^2 + c \sum x^3 = \sum xy \quad (26)$$

$$a \sum x^2 + b \sum x^3 + c \sum x^4 = \sum x^2 y \quad (27)$$

Here, the number of observations are $n = 7$ and relevant table is

Continued Unit - II Problems on Curve Fitting of a second degree Parabola ...

x	y	x^2	x^3	x^4	xy
1	2.3	1	1	1	2.3
2	5.2	4	8	16	10.4
3	9.7	9	27	81	29.1
4	16.5	16	64	256	66.0
5	29.4	25	125	625	147.0
6	35.5	36	216	1296	213.0
7	54.4	49	343	2401	380.8
$\sum x = 28$	$\sum y = 153$	$\sum x^2 = 140$	$\sum x^3 = 784$	$\sum x^4 = 4676$	$\sum xy = 848.6$

Substituting all the values in the normal equations, we get the following equations

$$7a + 28b + 140c = 153 \quad (28)$$

$$28a + 140b + 784c = 848.6 \quad (29)$$

$$140a + 784b + 4676c = 5053 \quad (30)$$

By solving the equations (30), (31) and (32) using elimination method, we get $a = 2.3731$, $b = -1.094$ and $c = 1.193$. Substituting the values of a , b and c in equation (26). Hence the required equation of the parabola for the given data is

$$y = 2.3731 - 1.094x + 1.193x^2$$

Continued Unit - II Problems on Curve Fitting of a second degree Parabola ...

Problem - 02: Find a law of the form $y = a + bx^2$ by the method of least - squares to the following data.

x	10	20	30	40	50
y	8	10	15	21	30

Solution: By data, we need to find the second degree parabola of the form

$$y = a + bx^2 \quad (31)$$

We know that, the normal equations for the given parabola are

$$na + b \sum x^2 = \sum y \quad (32)$$

$$a \sum x^2 + b \sum x^4 = \sum x^2 y \quad (33)$$

Here, the number of observations are $n = 5$ and relevant table is

x	y	x^2	x^4	$x^2 y$
10	8			
20	10			
30	15			
40	21			
50	30			
$\sum x =$	$\sum y =$	$\sum x^2 =$	$\sum x^4 =$	$\sum x^2 y =$

Continued Unit - II on Correlation ...

Correlation: Consider any two independent magnitudes x and y , then the co - variations between two independent magnitudes is known as **Correlation**. The relation between two variables x and y are decreases (or) increases in one of them corresponding to decreases (or) increases in the other then variables are called positively correlated. The relation between two variables x and y are decreases (or) increases in one of them corresponding to increases (or) decreases in the other then variables are called negatively correlated.

Correlation Coefficient (OR) Co - efficient of Correlation: The numerical measure of correlation between two variables x and y is called as coefficient of correlation. The coefficient of correlation is denoted by r and is given by

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{n\sigma_x\sigma_y}$$

Where, $\bar{x}$ is the mean value of x , $\bar{y}$ is the mean value of y , σ_x is the standard deviation of x and σ_y is the standard deviation of y .

Let $X = x - \bar{x}$ and $Y = y - \bar{y}$, then the standard deviations of x and y are given by

$$\sigma_x = \sqrt{\sum \frac{(x - \bar{x})^2}{n}} = \sqrt{\sum \frac{X^2}{n}}$$

$$\sigma_y = \sqrt{\sum \frac{(y - \bar{y})^2}{n}} = \sqrt{\sum \frac{Y^2}{n}}$$

Substituting all these in the above formula and simplify we get, an alternative formula for the coefficient of correlation

$$r = \frac{\sum XY}{(\sqrt{\sum X^2})(\sqrt{\sum Y^2})}$$

Theorem: Prove that the co-efficient of correlation r lies between -1 and $+1$ (i.e., $-1 \leq r \leq +1$).

Proof: We know that, by Cauchy - Schwartz inequality states that, if A and B are two set of values then

$$(\sum AB)^2 \leq (\sum A^2)(\sum B^2)$$

Therefore,

$$\frac{(\sum AB)^2}{(\sum A^2)(\sum B^2)} \leq 1 \quad (34)$$

Continued Unit - II on Correlation ...

We have, the co-efficient of correlation r is

$$r = \frac{\sum XY}{(\sqrt{\sum X^2})(\sqrt{\sum Y^2})}$$

squaring on both sides, we get

$$r^2 = \frac{(\sum XY)^2}{(\sum X^2)(\sum Y^2)} \quad (35)$$

Now, comparing the equations (40) and (41) we get

$$r^2 \leq 1$$

$$\rightarrow r \leq \pm\sqrt{1}, r \leq \pm 1$$

$$-1 \leq r \leq +1$$

Hence, the co-efficient of correlation r lies between -1 and $+1$.

Theorem: If $z = ax + by$ and r is the co-efficient of correlation between x and y , then prove that

$$\sigma_z^2 = a^2\sigma_x^2 + b^2\sigma_y^2 + 2abr\sigma_x\sigma_y.$$

Hence show that

$$r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}.$$

Continued Unit - II on Correlation ...

Proof: Given

$$z = ax + by$$

Taking summation on both sides and dividing by n , we get

$$\frac{\sum z}{n} = \frac{\sum(ax + by)}{n}$$

$$\frac{\sum z}{n} = a\left(\frac{\sum x}{n}\right) + b\left(\frac{\sum y}{n}\right)$$
$$\bar{z} = a\bar{x} + b\bar{y}$$

Now, consider

$$(z - \bar{z})^2 = [(ax + by) - (a\bar{x} + b\bar{y})]^2$$

$$(z - \bar{z})^2 = [a(x - \bar{x}) + b(y - \bar{y})]^2$$

Expanding RHS by using $(a + b)^2$ expansion

$$(z - \bar{z})^2 = a^2(x - \bar{x})^2 + b^2(y - \bar{y})^2 + 2ab(x - \bar{x})(y - \bar{y})$$

Taking the summation and dividing by n on both sides

Continued Unit - II on Correlation ...

$$\frac{\sum(z - \bar{z})^2}{n} = a^2 \frac{\sum(x - \bar{x})^2}{n} + b^2 \frac{\sum(y - \bar{y})^2}{n} + 2ab \frac{\sum(x - \bar{x})(y - \bar{y})}{n} \quad (36)$$

We know that, $\sigma_x^2 = \frac{\sum(x - \bar{x})^2}{n}$, $\sigma_y^2 = \frac{\sum(y - \bar{y})^2}{n}$, $\sigma_z^2 = \frac{\sum(z - \bar{z})^2}{n}$ and the co-efficient of correlation r

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{n\sigma_x\sigma_y}$$

From this we have,

$$\frac{\sum(x - \bar{x})(y - \bar{y})}{n} = r\sigma_x\sigma_y$$

Substituting all these in equation(42), we get

$$\sigma_z^2 = a^2\sigma_x^2 + b^2\sigma_y^2 + 2abr\sigma_x\sigma_y$$

Hence it is proved the first part and to deduce the alternative formula for co-efficient of correlation r , substitute $a = 1$ and $b = -1 \rightarrow z = x - y$ in above equation, we get

$$\begin{aligned}\sigma_{x-y}^2 &= (1)^2\sigma_x^2 + (-1)^2\sigma_y^2 + 2(1)(-1)r\sigma_x\sigma_y \\ \sigma_{x-y}^2 &= \sigma_x^2 + \sigma_y^2 - 2r\sigma_x\sigma_y\end{aligned}$$

Therefore,

$$r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$$

Continued Unit - II on Correlation ...

Working procedure to find the correlation co - efficient

Step - 01: Write the formula for correlation co - efficient r

Formula - 1:

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{n\sigma_x\sigma_y}$$

Formula - 2:

$$r = \frac{\sum XY}{(\sqrt{\sum X^2})(\sqrt{\sum Y^2})}$$

Here, $X = x - \bar{x}$ and $Y = y - \bar{y}$

Formula - 3:

$$r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$$

Step - 02: Construct the table from the given data according to the formula.

Step - 03: Substituting all the values in formula and simplify, we get the co - efficient of correlation for the given data.

Continued Unit - II Problems on Correlation ...

Problems:

Problem - 01: Find the correlation co-efficient for the following data.

x	1	2	3	4	5
y	2	5	3	8	7

Solution: We have, correlation co-efficient r is

$$r = \frac{\sum XY}{(\sqrt{\sum X^2})(\sqrt{\sum Y^2})}$$

$$\text{Mean of } x = \bar{x} = \frac{\sum x}{n} = \frac{1+2+3+4+5}{5} = 3$$

$$\text{Mean of } y = \bar{y} = \frac{\sum y}{n} = \frac{2+5+3+8+7}{5} = 5$$

Now construct the table

Continued Unit - II Problems on Correlation ...

x	y	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2	XY
1	2	-2	-3	4	9	6
2	5	-1	0	1	0	0
3	3	0	-2	0	4	0
4	8	1	3	1	9	3
5	7	2	2	4	4	4
				$\sum X^2 = 10$	$\sum Y^2 = 26$	$\sum XY = 13$

Substitute all the table values in above formula

$$r = \frac{13}{(\sqrt{10})(\sqrt{26})} = \frac{13}{16.1245} = 0.806$$

Problem - 02: Find the correlation co - efficient for the following data.

x	10	14	18	22	26	30
y	18	12	24	6	30	36

Solution: We have, correlation co - efficient r is

Problem - 03: Find the correlation co - efficient for the data given below.

x	23	27	28	29	30	31	33	35	36	39
y	18	22	23	24	25	26	28	29	30	32

Solution: We have, correlation co - efficient r is

Continued Unit - II Regression ...

Regression: Suppose $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$, are the given n pairs of values of two variables x and y , then the process of estimating one variable y in terms of other variable x is known as regression. If finding a relation like $y = f(x)$ (*i.e.* y is dependent variable) in terms of x (*i.e.* x is independent variable) is called the equation of regression line of y on x . There are two types of regressions.

(i) Equation of regression line of y on x .

(ii) Equation of regression line of x on y .

Obtain the equation of regression line of y on x :

Solution: Consider a set of n value of two variables x and y are

$(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$,. Let $y = ax + b$ be the equation of the regression line of y on x for a given set values of x and y . We know that, the normal equations to the above straight line are

$$a \sum x^2 + b \sum x = \sum xy$$

$$a \sum x + nb = \sum y$$

Dividing the second equation by n , we get

$$a \frac{\sum x}{n} + n \frac{b}{n} = \frac{\sum y}{n}$$

Continued Unit - II on Regression ...

Therefore,

$$a\bar{x} + b = \bar{y}$$

From the above equation we have observed that the equation of regression line is passes through $(\bar{x}, \bar{y})$ that is the mean values of x and y .

Also, we know that the equation of the straight line passing through the point (x_1, y_1) having the slope m is given by Therefore,

$$(y - y_1) = m(x - x_1)$$

Now, replace x_1 by $\bar{x}$ and y_1 by $\bar{y}$, we get

$$(y - \bar{y}) = m(x - \bar{x}) \quad (37)$$

$$Y = mX$$

where $Y = (y - \bar{y})$ and $X = (x - \bar{x})$ The normal equation to find the parameter m is

$$\sum XY = m \sum X^2$$

Hence, the parameter m is

$$m = \frac{\sum XY}{\sum X^2}$$

Continued Unit - IO on Regression ...

But we have the co-efficient of correlation r is

$$r = \frac{\sum XY}{n\sigma_x\sigma_y}$$

From this,

$$\sum XY = nr\sigma_x\sigma_y$$

Also, we have the variance of x is

$$\sigma_x^2 = \frac{\sum (x - \bar{x})^2}{n} = \frac{\sum X^2}{n}$$

$$\sigma_x^2 = \frac{\sum (x - \bar{x})^2}{n}$$

$$\sigma_x^2 = \frac{\sum X^2}{n}$$

→

$$\sum X^2 = n\sigma_x^2$$

Substituting $\sum XY$ and $\sum X^2$ in m

$$m = \frac{nr\sigma_x\sigma_y}{n\sigma_x^2}$$

Continued Unit - II on Regression ...

$$m = \frac{r\sigma_y}{\sigma_x}$$

Again substitute m in equation (43) we get

$$(y - \bar{y}) = \left(\frac{r\sigma_y}{\sigma_x}\right)(x - \bar{x})$$

Hence this is the equation of regression line of y on x and the coefficient of x is called the regression coefficient of y on x . The coefficient of regression of y on x is denoted by b_{yx} and is given by

$$b_{yx} = \frac{r\sigma_y}{\sigma_x}$$

Similarly, we can obtain the equation of regression line of x on y is

$$(x - \bar{x}) = \left(\frac{r\sigma_x}{\sigma_y}\right)(y - \bar{y})$$

The coefficient of y is called the coefficient of regression of x on y . The coefficient of regression of x on y is denoted by b_{xy} and is given by

$$b_{xy} = \frac{r\sigma_x}{\sigma_y}$$

Continued Unit - II on Regression ...

Problem: Prove that the correlation coefficient r is the geometric mean of regression coefficients.

Solution: We know that, the geometric mean between two numbers a and b is given by $\sqrt{aXb}$.

We have, the coefficients of regression of y on x and x on y are

$$b_{yx} = \frac{r\sigma_y}{\sigma_x}$$

and

$$b_{xy} = \frac{r\sigma_x}{\sigma_y}$$

Now, consider

$$b_{xy}Xb_{yx} = \left(\frac{r\sigma_x}{\sigma_y}\right)X\left(\frac{r\sigma_y}{\sigma_x}\right)$$

$$b_{xy}Xb_{yx} = r^2$$

$$r = \pm\sqrt{b_{xy}Xb_{yx}}$$

Hence, the coefficient of correlation r is a geometric mean between regression coefficients.

Continued Unit - II Problems on Regression ...

Note:

- (i) If both regression coefficients are positive then the correlation coefficient also positive.
- (ii) If both regression coefficients are negative then the correlation coefficient also negative.

Working procedure for finding equation of regression lines:

Step - 01: Write the formulas for the equation of regression lines.

Step - 02: Construct the table according to the formulas from the given data.

Step - 03: Substituting all the values in the formula and simplify, we get the equation for regression lines for the given data.

Step - 04: Suppose the equation of regression lines are given, solve the given equations find the mean values of x and y , also find the co-efficient of correlation r .

Problems:

Problem -01: Calculate the coefficient of correlation and obtain the equations of regression lines for the following data. Also estimate the value y when $x = 6.2$.

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

Solution: We know that, the coefficient of correlation r is

Continued Unit - II Problems on Regression ...

$$r = \frac{\sum XY}{(\sqrt{\sum X^2})(\sqrt{\sum Y^2})}$$

$$\text{Mean of } x = \bar{x} = \frac{\sum x}{n} = \frac{1+2+3+4+5+6+7+8+9}{9} = \frac{45}{9} = 5$$

$$\text{Mean of } y = \bar{y} = \frac{\sum y}{n} = \frac{9+8+10+12+11+13+14+16+15}{9} = \frac{108}{9} = 12$$

Now construct the table

x	$X = x - \bar{x}$	X^2	y	$Y = y - \bar{y}$	y^2	XY
1	-4	16	9	-3	9	12
2	-3	9	8	-4	16	12
3	-2	4	10	-2	4	4
4	-1	1	12	0	0	0
5	0	0	11	-1	1	0
6	1	1	13	1	1	1
7	2	4	14	2	4	4
8	3	9	16	4	16	12
9	4	16	15	3	9	12
		$\sum X^2 = 60$			$\sum Y^2 = 60$	$\sum XY = 57$

Continued Unit - II Problems on Regression ...

Substituting all the values in the above formula and simplify

$$r = \frac{57}{(\sqrt{60})(\sqrt{60})} = \frac{57}{60} = 0.95$$

We have, the equation of regression line of y on x is

$$(y - \bar{y}) = \left(\frac{r\sigma_y}{\sigma_x}\right)(x - \bar{x})$$

an the equation of regression line of x on y is

$$(x - \bar{x}) = \left(\frac{r\sigma_x}{\sigma_y}\right)(y - \bar{y})$$

→ The variance of x is

$$\sigma_x^2 = \frac{\sum X^2}{n} = \frac{60}{9} = 6.66667$$

and the standard deviation of x is

$$\sigma_x = \sqrt{\text{Variance}} = \sqrt{6.66667} = 2.58198$$

The variance of y is

$$\sigma_y^2 = \frac{\sum Y^2}{n} = \frac{60}{9} = 6.66667$$

Continued Unit - II Problems on Regression ...

also the standard deviation of y is

$$\sigma_y = \sqrt{\text{Variance}} = \sqrt{6.66667} = 2.58198$$

Again substitute these values in above formulas, we get the equation of regression line of y on x is

$$(y - 12) = \left(\frac{(0.95)(2.58198)}{(2.58198)} \right) (x - 5)$$

$$y - 12 = 0.95(x - 5)$$

$$y = 0.95x + 7.25$$

and the equation of regression line of x on y is

$$(x - 5) = \left(\frac{(0.95)(2.58198)}{(2.58198)} \right) (y - 12)$$

$$x - 5 = 0.95(y - 12)$$

$$x = 0.95y - 6.4$$

Hence, the value of y when $x = 6.2$ is

$$y = 0.95(6.2) + 7.25 = 13.14$$

Continued Unit - II Problems on Regression ...

Problem -02: Obtain the equations of regression lines and hence find the coefficient of correlation for the following data.

x	1	3	4	2	5	8	9	10	13	15
y	8	6	10	12	16	16	10	32	32	

Solution:

Problem -03: Find the coefficient of correlation and hence obtain the equations of regression lines for the following data.

x	1	2	3	4	5
y	2	5	3	8	7

Solution:

Problem -04: Given that $\bar{x} = 5$, $\bar{y} = 11$, $n = 9$, $\sum(x - \bar{x})^2 = 60$, $\sum(y - \bar{y})^2 = 70$, $\sum(x - \bar{x})(y - \bar{y}) = 62$, find r , b_{yx} and b_{xy} in the usual notations. Also, estimate the value of y when $x = 10$ and the value of x when $y = 10$.

Solution:

Continued Unit - II Problems on Regression ...

Problem -05: The equations of regression lines of two variables x and y are $y = 0.516x + 33.73$, $x = 0.512y + 32.52$. Find the correlation coefficient and the mean values of x and y .

Solution: Given that, the equations of regression lines are

$$y = 0.516x + 33.73$$

$$x = 0.512y + 32.52$$

We know that, the coefficient of correlation r is

$$r = \pm \sqrt{b_{xy}Xb_{yx}}$$

The coefficient of x in the first equation $= b_{yx} = +0.516$

The coefficient of y in the second equation $= b_{xy} = +0.512$

Therefore,

$$r = \pm \sqrt{(+0.516)X(+0.512)} = +0.51399$$

Also, we know that the equations of regression lines are pass through the point $(\bar{x}, \bar{y})$, replace x by $\bar{x}$, y by $\bar{y}$ the above are

$$\bar{y} = 0.516\bar{x} + 33.73$$

$$\bar{x} = 0.512\bar{y} + 32.52$$

Continued Unit - II Problems on Regression ...

Now, solving these equations by elimination method find $\bar{x}$ and $\bar{y}$, we get The mean value of $x = \bar{x} = 67.6667$

The mean value of $y = \bar{y} = 68.64605$

Problem -06: In a partially destroyed laboratory record of correlation data, the following results only are available. The variance of x is 9 and the equations of regression lines of two variables x and y are $4x - 5y + 33 = 0$, $20x - 9y = 107$. Find (i) the mean values of x and y , (ii) standard deviation of y and (iii) the correlation coefficient between x and y .

Solution: By data, the equations of regression lines are

$$4x - 5y + 33 = 0$$

$$4x - 5y = -33$$

$$20x - 9y = 107$$

Since the equations of regression lines are pass through the point $(\bar{x}, \bar{y})$, replace x by $\bar{x}$, y by $\bar{y}$ the above are

$$4\bar{x} - 5\bar{y} = -33$$

$$20\bar{x} - 9\bar{y} = 107$$

Now, solving these equations by elimination method find $\bar{x}$ and $\bar{y}$, we get

Continued Unit - II Problems on Regression ...

The mean value of $x = \bar{x} = 13$

The mean value of $y = \bar{y} = 17$

We know that, the coefficient of correlation r is

$$r = \pm \sqrt{b_{xy} X b_{yx}}$$

Rewrite the above equations

$$4x - 5y + 33 = 0$$

$$4x + 33 = 5y$$

$$5y = 4x + 33$$

dividing by 5 on both sides

$$y = \frac{4}{5}x + \frac{33}{5}$$

$\therefore$ the coefficient of x in the above equation $= b_{yx} = +\frac{4}{5}$

$$20x - 9y = 107$$

$$20x = 9y + 107$$

Dividing by 20

$$x = \frac{9}{20}y + \frac{107}{20}$$

Continued Unit - II Problems on Regression ...

$\therefore$ the coefficient of y in the above equation $= b_{xy} = +\frac{9}{20}$

Therefore,

$$r = \pm \sqrt{\left(+\frac{4}{5}\right)X\left(+\frac{9}{20}\right)} = +\frac{3}{5} = +0.6$$

We have, to find the standard deviation of $y = \sigma_y$, given that the variance of $= \sigma_x^2 = 9 \rightarrow \sigma_x = \sqrt{9} = 3$. We know that

$$b_{yx} = \frac{r\sigma_y}{\sigma_x}$$

$$\sigma_y = \frac{\sigma_x b_{yx}}{r}$$

$$\sigma_y = \frac{(3)\left(\frac{4}{5}\right)}{\left(\frac{3}{5}\right)} = 4$$

Problem -07: If θ is the angle between the equations of regression lines, then show that

$$\tan(\theta) = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1 - r^2}{r} \right).$$

Continued Unit - II Problems on Regression ...

Solution: We know that, the acute angle θ between two straight lines $y = m_1x + c_1$ and $y = m_2x + c_2$ then slope of the tangent is given by

$$\tan(\theta) = \frac{m_2 - m_1}{1 + m_1m_2} \quad (38)$$

We have, the equation of regression line of y on x is

$$(y - \bar{y}) = \left(\frac{r\sigma_y}{\sigma_x}\right)(x - \bar{x})$$

The slope of the tangent $= m_1 =$ coefficient of $x = \frac{r\sigma_y}{\sigma_x}$

Also, the equation of regression line of x on y is

$$(x - \bar{x}) = \left(\frac{r\sigma_x}{\sigma_y}\right)(y - \bar{y})$$

rewrite the equation we get

$$(y - \bar{y}) = \left(\frac{\sigma_y}{r\sigma_x}\right)(x - \bar{x})$$

The slope of the tangent $= m_2 =$ coefficient of $x = \frac{\sigma_y}{r\sigma_x}$

Continued Unit - II Problems on Regression ...

Substituting m_1 and m_2 in equation (44) and simplify we get

$$\tan(\theta) = \frac{\left(\frac{\sigma_y}{r\sigma_x}\right) - \left(\frac{r\sigma_y}{\sigma_x}\right)}{1 + \left(\frac{r\sigma_y}{\sigma_x}\right)\left(\frac{r\sigma_y}{\sigma_x}\right)}$$

Hence,

$$\tan(\theta) = \frac{\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1 - r^2}{r} \right).$$

Problem -08: In a bivariate distribution $\sigma_x = \sigma_y$ and the angle between the regression lines is $\tan^{-1}(3)$. Find the correlation coefficient.

Solution: By data $\sigma_x = \sigma_y$ and $\tan^{-1}(3)$

We know that, if θ is the angle between the equations of regression lines then

$$\tan(\theta) = \frac{\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1 - r^2}{r} \right)$$

$\rightarrow \theta = \tan^{-1}(3)$, $\tan(\theta) = 3$, now substitute $\tan(\theta) = 3$ and $\sigma_x = \sigma_y$ in above equation, we get

$$\begin{aligned} 3 &= \frac{\sigma_x\sigma_x}{\sigma_x^2 + \sigma_x^2} \left(\frac{1 - r^2}{r} \right) \\ 3 &= \frac{\sigma_x^2}{2\sigma_x^2} \left(\frac{1 - r^2}{r} \right) \end{aligned}$$

Continued Unit - II Problems on Regression ...

$$3 = \frac{1}{2} \left(\frac{1 - r^2}{r} \right)$$

$$6 = \frac{1 - r^2}{r}$$

$$6r = 1 - r^2$$

$$r^2 + 6r - 1 = 0$$

This is quadratic equation in r and the roots are

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r = \frac{-6 \pm \sqrt{(-6)^2 - 4(1)(-1)}}{2(1)}$$

$$r = \frac{-6 \pm \sqrt{36 + 4}}{2}$$

$$r = \frac{-6 \pm \sqrt{40}}{2}$$

$$r = \frac{-6 \pm 2\sqrt{10}}{2} = -3 \pm \sqrt{10}$$

$\therefore r = +0.1623 \in (-1, 1)$ and $r = -6.1623 \notin (-1, 1)$

Since, $-1 \leq r \leq 1$, $\therefore r = 0.1623$

Partial correlation:

The measure of the variation between the dependent variable which is depends on more than one independent variables $X_1, X_2, X_3, \dots, X_k$ is known as partial correlation.

Coefficient of partial correlation:

The numerical measure of the partial correlation between the dependent variable and one independent variable by treating the other variables are constant is called as the coefficient of partial correlation. Suppose Y, X_1 and X_2 are any three variables then the coefficient of partial correlation between first variable Y and the second variable X_1 by treating third variable X_2 as constant is denoted by $r_{12.3}$ and is given by

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\left(\sqrt{1 - r_{13}^2}\right)\left(\sqrt{1 - r_{23}^2}\right)}$$

Similarly, the coefficient of partial correlation between first variable Y and the third variable X_2 by treating second variable X_1 as constant is denoted by $r_{13.2}$ and is given by

$$r_{13.2} = \frac{r_{13} - r_{12}r_{23}}{\left(\sqrt{1 - r_{12}^2}\right)\left(\sqrt{1 - r_{23}^2}\right)}$$

and the coefficient of partial correlation between second variable X_1 and the third variable X_2 by treating first variable Y as constant is denoted by $r_{23.1}$ and is given by

$$r_{23.1} = \frac{r_{23} - r_{12}r_{13}}{\left(\sqrt{1 - r_{12}^2}\right)\left(\sqrt{1 - r_{13}^2}\right)}$$

Problem: On the basis of the following information compute the coefficient of the partial correlations $r_{12.3}$, $r_{13.2}$ and $r_{23.1}$.

Solution:

Continued Unit - II on Multiple Correlation ...

Multiple Correlation:

The measure of co - variation between the dependent variable y which is depends on more than one independent variables $X_1, X_2, X_3, \dots, X_k$ is known as the multiple correlation.

Co - efficient of Multiple Correlation: The numerical measure of the multiple correlation between the dependent variable y which is depends on more than one independent variables $X_1, X_2, X_3, \dots, X_k$ is called as the coefficient of multiple correlation. Suppose there are three variables Y, X_1 and X_2 then the coefficient of multiple correlation between Y which depends on two independent variables X_1 and X_2 for the given data is denoted by $R_{1.23}$ and is given by

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}}$$

Similarly,

$$R_{2.13} = \sqrt{\frac{r_{12}^2 + r_{23}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{12}^2}}$$

and

$$R_{3.12} = \sqrt{\frac{r_{13}^2 + r_{23}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{12}^2}}$$

Continued Unit - II on Multiple Correlation ...

where

r_{12} is the coefficient of partial correlation between first variable Y and the second variable X_1 .

r_{13} is the coefficient of partial correlation between the first variable Y and the third variable X_2 .

r_{23} is the coefficient of partial correlation between the second variable X_1 and the third variable X_2 .

Note:

- 1 The coefficient of multiple correlation R takes the values between 0 and 1.
- 2 The coefficient of multiple correlation R takes the higher values indicates higher predictability of the dependent variable from the independent variables.
- 3 The coefficient of multiple correlation R takes the value of 1 indicating that the predictions are exactly correct.
- 4 The coefficient of multiple correlation R takes the value of 0 indicating that no linear relationship between the variables and it is possible that a non - linear relationship may exists.

Problem: The following coefficient of the partial correlation are given $r_{12} = 0.98$, $r_{13} = 0.44$ and $r_{23} = 0.54$. Calculate the coefficient of the multiple correlation treating first variable as dependent variable and second and third variables as independent variables and hence compute $R_{2.13}$ and $R_{3.12}$.

Solution:

Continued Unit - II on Multiple Regression ...

Multiple Regression:

It is known that, in agriculture that, the crop yield (Y) not only depends on the amount of rain fall(X_1) but also on the amount of fertilizers (X_2) applied, pesticides (X_3) used, quality of seeds (X_4) and quality of soil (X_5) etc.. Thus in multiple regression, the dependent variable Y is a function of more than one independent variables X_1, X_2, X_3, X_4 and X_k is

$$Y = f(X_1, X_2, X_3, X_4, \dots, X_k)$$

In multiple non - linear regression, f is non linear and in multiple linear regression f is linear. that is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

Response surface analysis deals with statistical methods of prediction and optimization.

Linear Multiple Regression:

Suppose the dependent variable Y is depends on two independent variables X_1 and X_2 of the form

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

Then the linear multiple regression is to determine the coefficients β_0, β_1 and β_2 for the given set of data.

Continued Unit - II on Multiple Regression ...

This problem is essentially to fit regression plane $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$ to a given set of N triples (X_{1i}, X_{2i}, Y_i) , where $i = 1, 2, 3, \dots, N$. To estimate the coefficients $\beta_0, \beta_1, \beta_2$, by applying the method of least square technique.

Consider set of N triples (X_{1i}, X_{2i}, Y_i) in a space. Our aim is to find out an equation of the plane $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$ by least square technique.

Let the residual $R = Y_i - (\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i})$ is the difference between observed value and the estimates values of Y . By the method of least square, we need to find out an approximate values of the parameters β_0, β_1 and β_2 from the given set of data such that the sum of the square of the residual is minimum.

$$S = \sum_{i=1}^N [Y_i - (\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i})]^2$$

Clearly S is a function of β_0, β_1 and β_2 , the necessary conditions for which S is to be minimum are $\frac{\partial S}{\partial \beta_0} = 0$, $\frac{\partial S}{\partial \beta_1} = 0$ and $\frac{\partial S}{\partial \beta_2} = 0$.

$$\frac{\partial S}{\partial \beta_0} = 0$$

$\Rightarrow$

$$\sum 2[Y_i - (\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i})](-1) = 0$$

Continued Unit - II on Multiple Regression ...

$$\sum Y_i = N\beta_0 + \beta_1 \sum X_{1i} + \beta_2 \sum X_{2i}$$

Similarly,

$$\frac{\partial S}{\partial \beta_1} = 0$$

$\Rightarrow$

$$\begin{aligned} \sum 2[Y_i - (\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i})](-X_{1i}) &= 0 \\ \sum X_{1i} Y_i &= \beta_0 \sum X_{1i} + \beta_1 \sum X_{1i}^2 + \beta_2 \sum X_{1i} X_{2i} \end{aligned}$$

and

$$\frac{\partial S}{\partial \beta_2} = 0$$

$\Rightarrow$

$$\begin{aligned} \sum 2[Y_i - (\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i})](-X_{2i}) &= 0 \\ \sum X_{2i} Y_i &= \beta_0 \sum X_{2i} + \beta_1 \sum X_{1i} X_{2i} + \beta_2 \sum X_{2i}^2 \end{aligned}$$

These three equations together refer to as normal equations to fit a regression plane
 $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$.

Note:

$$\bar{x}_1 = \frac{\sum X_{1i}}{N}, \bar{x}_2 = \frac{\sum X_{2i}}{N}, \bar{Y} = \frac{\sum Y_i}{N}$$

For problems:

To fit the plane $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$ for the given set of data. The normal equations are

$$\sum Y = n\beta_0 + \beta_1 \sum X_1 + \beta_2 \sum X_2$$

$$\sum X_1 Y = \beta_0 \sum X_1 + \beta_1 \sum X_1^2 + \beta_2 \sum X_1 X_2$$

$$\sum X_2 Y = \beta_0 \sum X_2 + \beta_1 \sum X_1 X_2 + \beta_2 \sum X_2^2$$

Continued Unit - II Problems on Multiple Regression ...

Problem - 01: Fit a regression plane to estimate β_0, β_1 and β_2 to the following data of a transport company on the weight of 6 shipments, the distances they were moved and the damage of the goods that was increased. Estimate the damage where a shipment of 3700 kg is moved a distance of 260 km.

Weight (x_1) (1000 kg)	4.0	3.0	1.6	1.2	3.4	4.8
Distance(x_2) (100 km)	1.5	2.2	1.0	2.0	.0.8	1.6
Damage (y) Re	160	112	69	90	123	186

Solution: The dependent variable y be the damage and the two independent variables are weight x_1 and distance x_2 . Thus assume the equation of regression plane as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \quad (39)$$

where β_0, β_1 and β_2 are the parameters to be determine and the normal equations for fitting the regression plane (39) are

$$\begin{aligned}\sum y &= n\beta_0 + \beta_1 \sum x_1 + \beta_2 \sum x_2 \\ \sum x_1 y &= \beta_0 \sum x_1 + \beta_1 \sum x_1^2 + \beta_2 \sum x_1 x_2 \\ \sum x_2 y &= \beta_0 \sum x_2 + \beta_1 \sum x_1 x_2 + \beta_2 \sum x_2^2\end{aligned}$$

Continued Unit - II Problems on Multiple Regression ...

Construct table according to the above normal equations

x_1	x_2	y	x_1^2	x_2^2	x_1x_2
4.0	1.5	160	16	2.25	6.0
3.0	2.2	112	9	4.84	6.6
1.6	1.0	69	2.56	1.0	1.6
1.2	2.0	90	1.44	2.4	2.4
3.4	0.8	123	11.56	2.72	2.72
4.8	1.6	186	23.04	7.68	7.68
$\sum x_1 = 18$	$\sum x_2 = 9.1$	$\sum y = 740$	$\sum x_1^2 = 63.6$	$\sum x_2^2 = 15.29$	$\sum x_1x_2 = 27$

Substituting these values in the above normal equations, we get

$$6\beta_0 + 18\beta_1 + 9.1\beta_2 = 740$$

$$18\beta_0 + 63.6\beta_1 + 27\beta_2 = 250.54$$

$$9.1\beta_0 + 27\beta_1 + 15.29\beta_2 = 1131.4$$

By solving above system of equations using elimination method, we get $\beta_0 = 14.56$, $\beta_1 = 30.109$ and $\beta_2 = 12.16$ and substituting these value in the above regression plane

$$y = 14.56 + 30.109x_1 + 12.16x_2 \quad (40)$$

END – OF – THE – SECOND – UNIT – II